Evaluation Sets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected evaluation record with a ranking extension.

Explain a case, trace rates, audit coverage and design a fully specified check.

Keep one habit: Retain source versions, expected behavior, observations, subgroup denominators and exposure history.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Quality Rubrics

Block to Python Bridge

Conversation State and Memory

W1 · Explain an evaluation case

> Give input, permitted evidence, expected behavior and exposure.

Your evidence record

- Ten-case W2:6 answerable with 5 correct, 4 missing-evidence with 3 appropriate fallbacks; rates 5/6 and 3/4, pooled 8/10 when each case has one required outcome. Separate pictured record:2/3answering,3/3fallback,5/6overall; adding3fallback passes gives 8/9 overall but answering still 2/3. These supplied outcomes are fictional, not live model results. Preserve exposure and case groups.

- Reference for this exercise: open this week's student lesson, "Evaluation Sets", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the ten-case rates

> Use separate category denominators before pooling.

Your evidence record

- Ten-case W2:6 answerable with 5 correct, 4 missing-evidence with 3 appropriate fallbacks; rates 5/6 and 3/4, pooled 8/10 when each case has one required outcome. Separate pictured record:2/3answering,3/3fallback,5/6overall; adding3fallback passes gives 8/9 overall but answering still 2/3. These supplied outcomes are fictional, not live model results. Preserve exposure and case groups.

- Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit coverage and tuning limits

> Visible reviewed cases do not establish final generalisation.

Your evidence record

- Ten-case W2:6 answerable with 5 correct, 4 missing-evidence with 3 appropriate fallbacks; rates 5/6 and 3/4, pooled 8/10 when each case has one required outcome. Separate pictured record:2/3answering,3/3fallback,5/6overall; adding3fallback passes gives 8/9 overall but answering still 2/3. These supplied outcomes are fictional, not live model results. Preserve exposure and case groups.

- A report omits this qualification: Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing evaluation case

> Supply all records and expected behavior without invented observations.

Your evidence record

- Ten-case W2:6 answerable with 5 correct, 4 missing-evidence with 3 appropriate fallbacks; rates 5/6 and 3/4, pooled 8/10 when each case has one required outcome. Separate pictured record:2/3answering,3/3fallback,5/6overall; adding3fallback passes gives 8/9 overall but answering still 2/3. These supplied outcomes are fictional, not live model results. Preserve exposure and case groups.

- Reference for this exercise: open this week’s student lesson, “Evaluation Sets”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Design checks for the fresh club records independently

> Keep the first attempt before hints.

Your evidence record

- Use only the supplied club notes and questions. The added unversioned note has no supplied precedence or authority rule. Do not assume a later-looking note overrides another, or invent a club leader. Design checks and expected responses; no actual system outcomes are supplied.

- Approved notes: Robotics Tuesday Room2; Art Wednesday Room5. Requests: “Where does the club meet?” and “Who leads drama?” An unversioned added note says Robotics Room3. Design checks and safe responses.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

My independent explanation, trace and limits

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.
- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

-
  - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Trace the worked case for System **evaluation set**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for System **evaluation set**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A test set contains ten easy questions with direct answers in the notes. Add three different kinds of cases that probe fallback behaviour and define expected outcomes.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Ten evaluation questions are easy definitions; none lacks a source.

What is untested?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

You improve a prompt after reading every test question. Can those questions still provide an untouched final evaluation? Propose a repair.

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 26 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A test set contains ten easy questions with direct answers in the notes. Add three different kinds of cases that probe fallback behaviour and define expected outcomes.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier twelve-case rate calculation

REFERENCE · SECTION 15

Additional practice · Earlier twelve-case rate calculation

> This retained task differs from the selected club-record assessment.

Earlier task and response

- Twelve cases: eight answerable, four unanswerable. Correct answers six; appropriate abstentions two. Compute separate and combined pass rates.

> **My reasoning and evidence**

▼ Extension · Ranking and retrieval evaluation — W1 Explain the mechanism

REFERENCE · SECTION 17

Extension · Ranking and retrieval evaluation — W1 Explain the mechanism

> Keep the cutoff, eligible judged set and denominator beside the task.

Complete ranking procedure

Ranking extension: known relevant eligible documents A,C,E; exactly three distinct top results A,B,C. Relevant returned 2 gives precision@3=2/3 and recall@3=2/3. Count document IDs once; a duplicate-filled list does not meet this exact-three-distinct-results contract. If the full relevant set is unknown, recall is unavailable.

If no relevant eligible documents exist, state denominator 0 and use an explicit unavailable policy rather than divide by 0. Independent ranking tasks use their own supplied sets.

Extension question and response

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

▼ **Extension · Ranking and retrieval evaluation — W2 Trace the worked case**

REFERENCE · SECTION 18

Extension · Ranking and retrieval evaluation — W2 Trace the worked case

> Keep the cutoff, eligible judged set and denominator beside the task.

Complete ranking procedure

Ranking extension: known relevant eligible documents A,C,E; exactly three distinct top results A,B,C. Relevant returned 2 gives precision@3=2/3 and recall@3=2/3. Count document IDs once; a duplicate-filled list does not meet this exact-three-distinct-results contract. If the full relevant set is unknown, recall is unavailable.

If no relevant eligible documents exist, state denominator 0 and use an explicit unavailable policy rather than divide by 0. Independent ranking tasks use their own supplied sets.

Extension question and response

- Relevant documents are A, C, E. Top three results A, B, C contain two relevant items. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

▼ **Extension · Ranking and retrieval evaluation — W3 Audit the limit**

REFERENCE · SECTION 19

Extension · Ranking and retrieval evaluation — W3 Audit the limit

> Keep the cutoff, eligible judged set and denominator beside the task.

Complete ranking procedure

Ranking extension: known relevant eligible documents A,C,E; exactly three distinct top results A,B,C. Relevant returned 2 gives precision@3=2/3 and recall@3=2/3. Count document IDs once; a duplicate-filled list does not meet this exact-three-distinct-results contract. If the full

relevant set is unknown, recall is unavailable.

If no relevant eligible documents exist, state denominator 0 and use an explicit unavailable policy rather than divide by 0. Independent ranking tasks use their own supplied sets.

Extension question and response

- A report omits this qualification: Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

▼ Extension · Ranking and retrieval evaluation — W4 Apply a new case independently

REFERENCE · SECTION 20

Extension · Ranking and retrieval evaluation — W4 Apply a new case independently

> Retained extension task with its own teacher key and response.

Extension question and record

- Relevant set A, D, F, G; top three A, B, D. Compute precision and recall. Explain which omitted records affect recall.

My reasoning and evidence

▼ Extension · Ranking and retrieval evaluation — W5 Design a revealing follow-up

REFERENCE · SECTION 21

Extension · Ranking and retrieval evaluation — W5 Design a revealing follow-up

> Keep the cutoff, eligible judged set and denominator beside the task.

Complete ranking procedure

Ranking extension: known relevant eligible documents A,C,E; exactly three distinct top results A,B,C. Relevant returned 2 gives precision@3=2/3 and recall@3=2/3. Count document IDs once; a duplicate-filled list does not meet this exact-three-distinct-results contract. If the full relevant set is unknown, recall is unavailable.

If no relevant eligible documents exist, state denominator 0 and use an explicit unavailable policy rather than divide by 0. Independent ranking tasks use their own supplied sets.

Extension question and response

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Design an **evaluation set** that checks answering and appropriate uncertainty separately.

Six toy questions: three have source answers and three do not. Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/26

Evaluation Sets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Separate cases

Keep answerable and missing-evidence cases as different groups.



2. Calculate rates

Answering: 2/3. Appropriate uncertainty: 3/3.



3. Report total

Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed.

CHANGE ONE THING

Add three missing-evidence cases that all pass.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Verified Study Buddy

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- Design an **evaluation set** that checks answering and appropriate uncertainty separately.
- Which labelled step matches this detail? Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed.
- 1. Separate cases
- 2. Calculate rates
- 3. Report total

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Report total Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. Read the picture in order.

  Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after trying**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Add three missing-**evidence** cases that all pass. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Overall becomes 8/9, but supported answering remains 2/3.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Overall becomes 8/9, but supported answering remains 2/3. Changing the case mixture raises the pooled rate without improving any existing answerable result.

> **My selected conclusion, deciding fact and explanation**

▼ **Check changed case**

Repair a missing building block

Repair and keep evidence

Review Quality Rubrics (ai-3/25). Goal: Judge an output using separate observable rubric criteria. Short practice: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

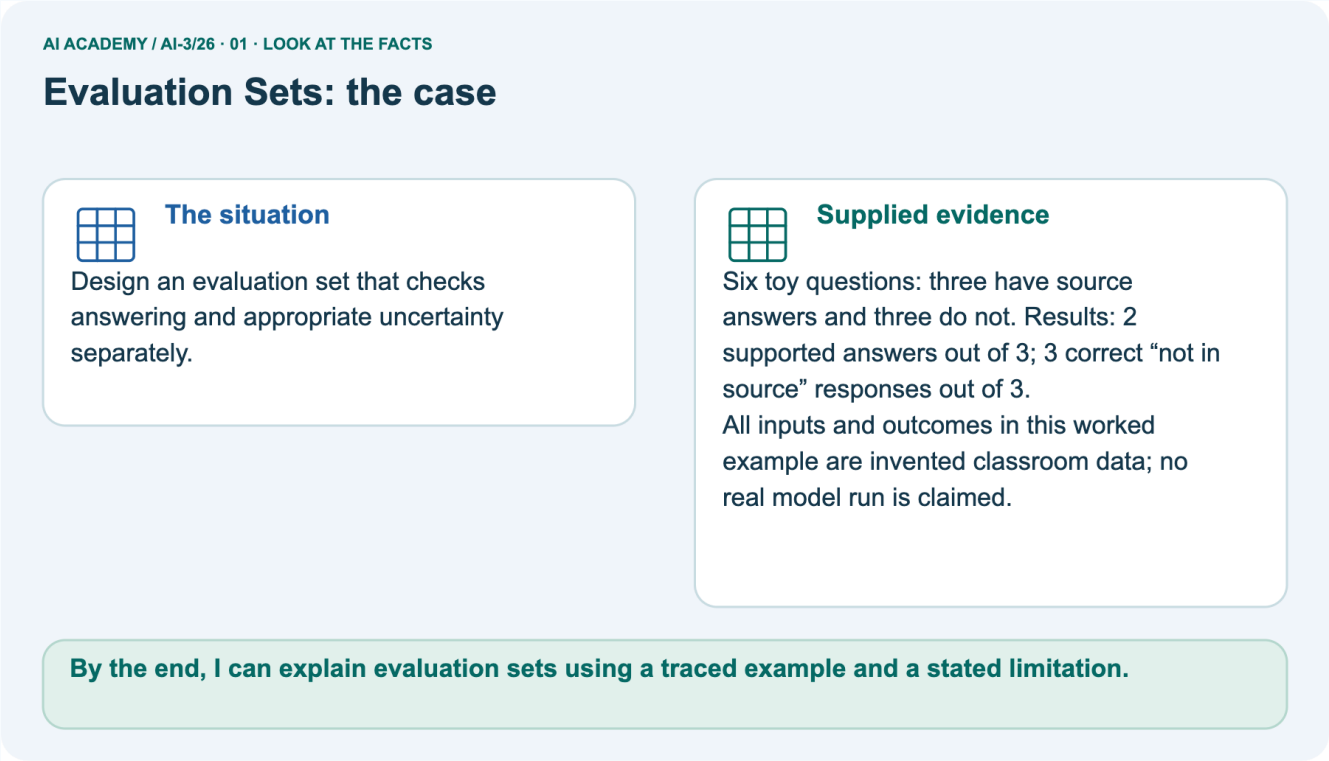
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. Evaluation Sets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Design an **evaluation set** that checks answering and appropriate uncertainty separately. Supplied facts: Six toy questions: three have source answers and three do not.

Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

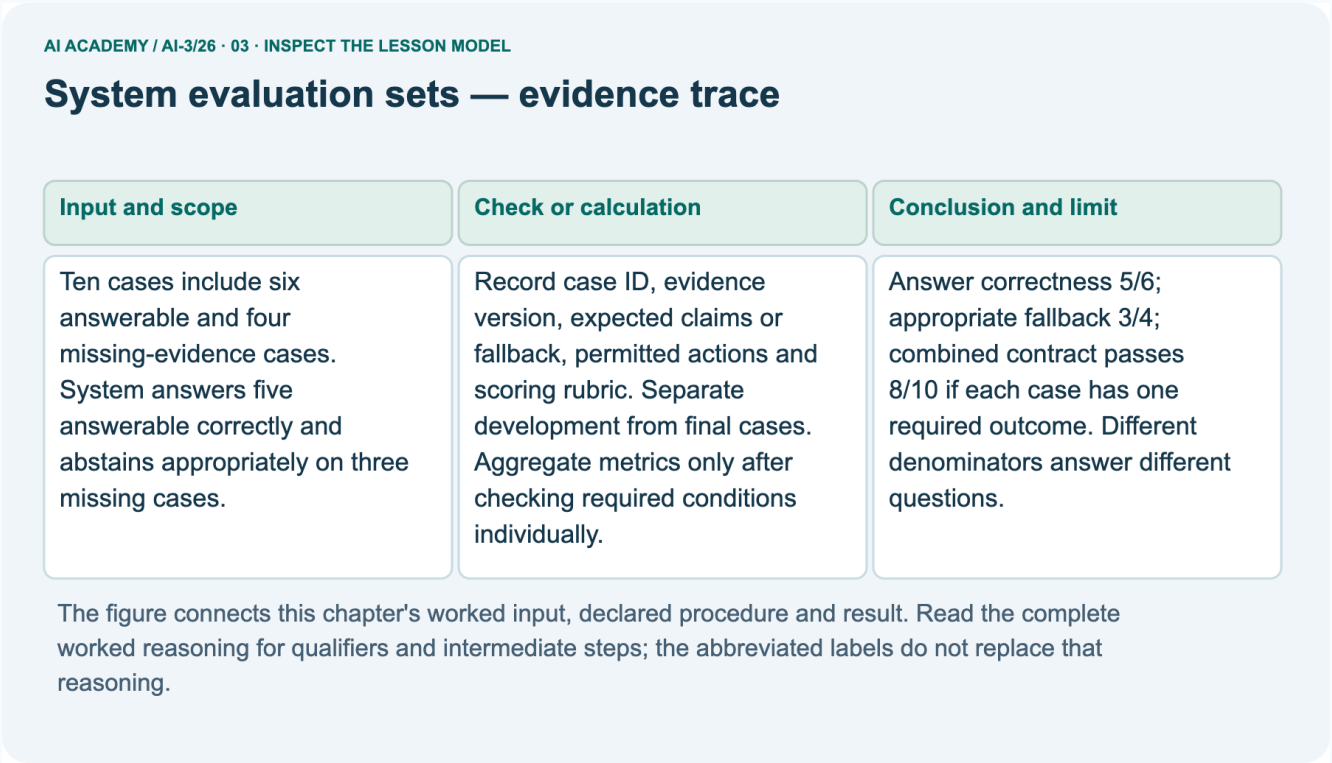
Separate cases: Keep answerable and missing-**evidence** cases as different groups. → Calculate rates: Answering: 2/3.

Appropriate uncertainty: 3/3. → Report total: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. System evaluation sets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

System **evaluation set**s — **evidence** trace. Ten cases include six answerable and four missing-**evidence** cases.

System answers five answerable correctly and abstains appropriately on three missing cases. | Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

Separate development from final cases.

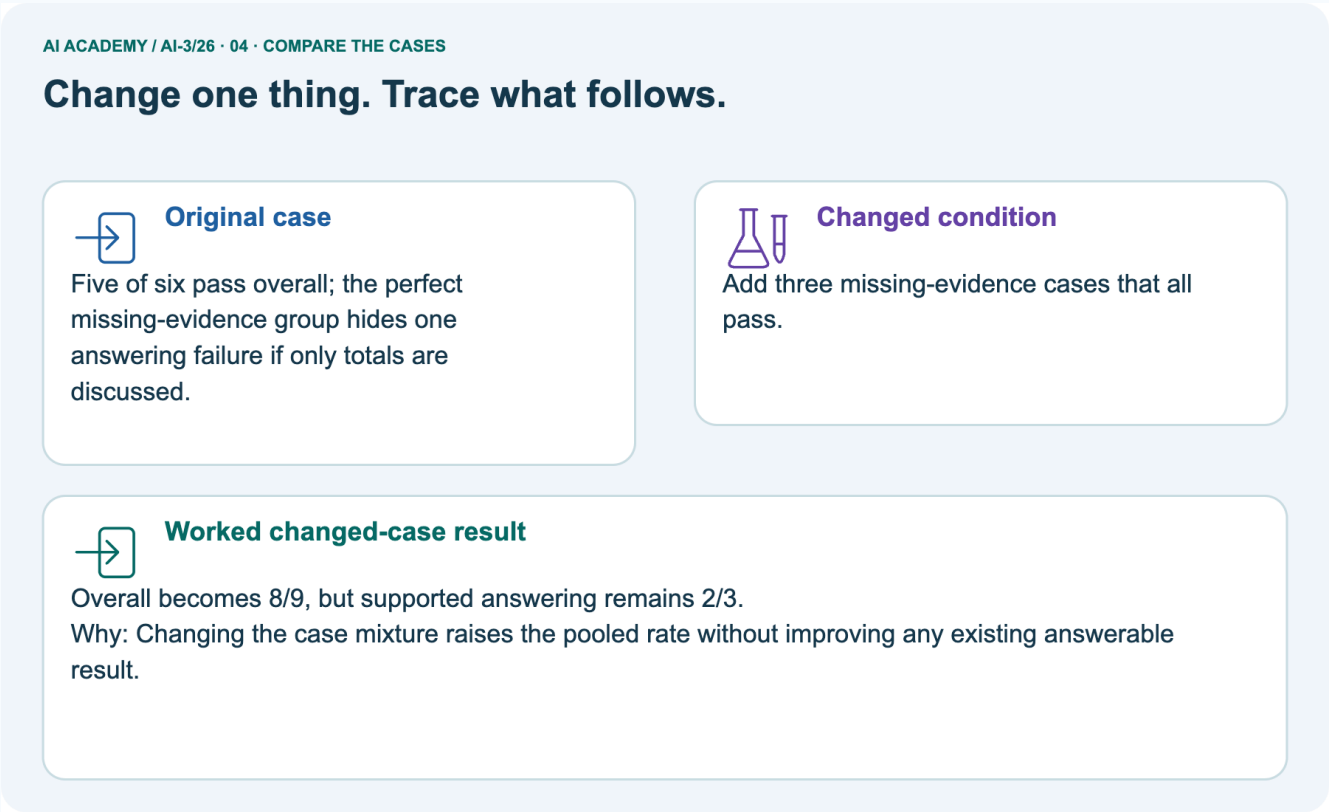
Aggregate metrics only after checking required conditions individually. | Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome.

Different denominators answer different questions.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

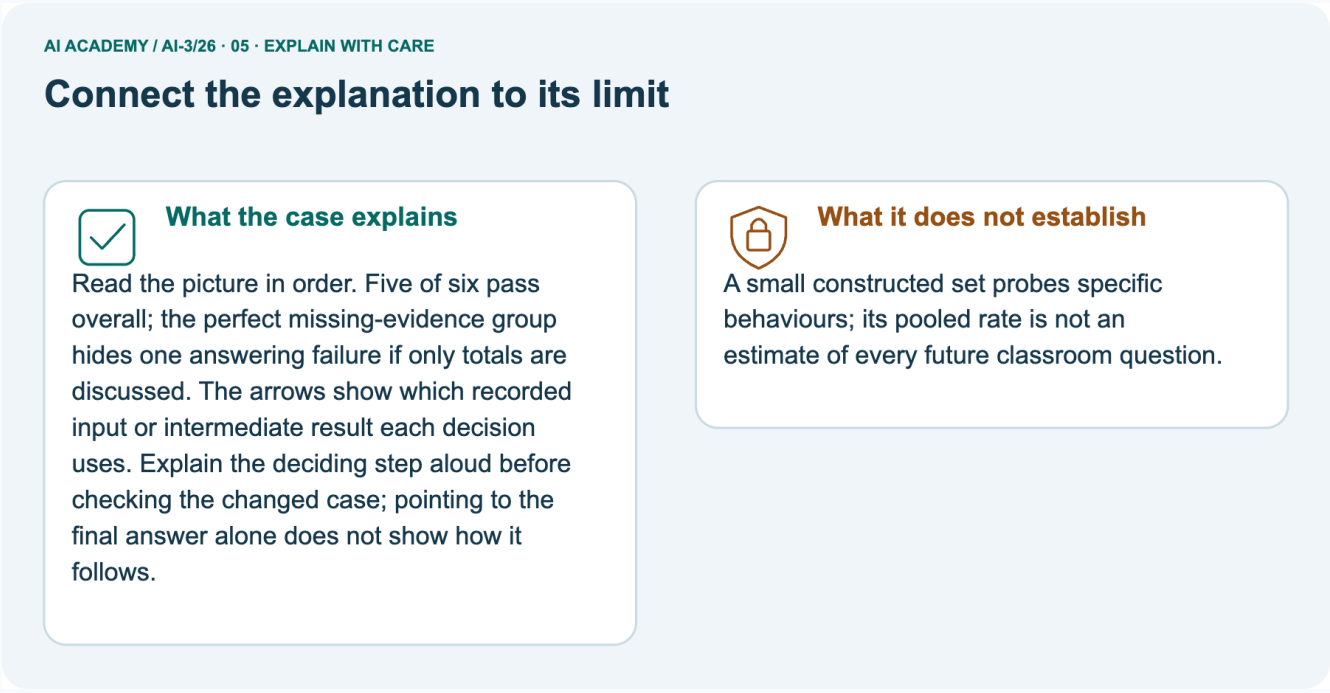
Original case: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. Changed condition: Add three missing-**evidence** cases that all pass.

Expected result: Overall becomes 8/9, but supported answering remains 2/3. Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

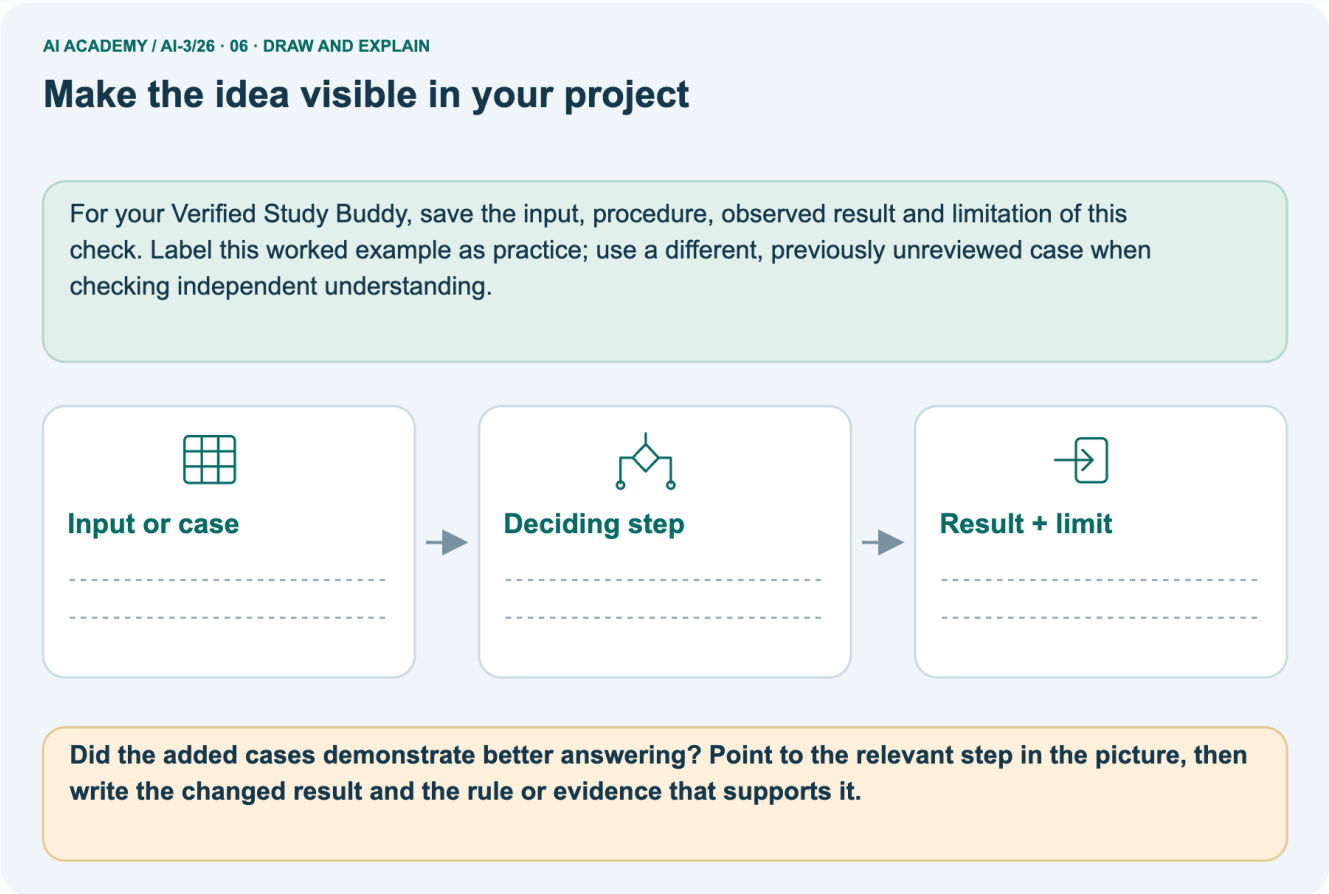
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Evaluation Sets: the case

02. Follow the deciding steps

03. System evaluation sets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.